

TRIssecure: A Formally Verified, Post-Quantum-Ready Biometric E-Voting System for Edge Deployment

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Abstract—Electronic voting systems must simultaneously satisfy strong voter authentication, tamper-evident vote integrity, end-to-end voter verifiability, and practical deployment constraints on resource-limited hardware. Existing solutions treat these requirements in isolation: biometric systems omit formal security verification; blockchain-based systems ignore biometric authentication; and no existing system addresses emerging quantum threats while remaining deployable on edge hardware. TRIssecure V2 is a three-factor biometric e-voting terminal that jointly satisfies all four requirements on a commodity Raspberry Pi 4 (ARM64, sub-\$100) without requiring server-grade computing or persistent cloud connectivity. It integrates an ISO/IEC 19795-1 compliant biometric pipeline with Bayesian score-level fusion of face and NFC card modalities, ISO/IEC 30107-3 presentation attack detection (MiniFASNetV2 liveness), a hybrid post-quantum cryptographic layer (Kyber-768+X25519 KEM and Dilithium-3+Ed25519 signatures per NIST FIPS 203/204), and a formally verified five-stage DFA authentication protocol with six proved security properties. Simulation-based evaluation achieves EER = 1.55%, AUC = 0.9991, TAR@1%FAR = 97.7%, and ACER = 4.25%. On-device benchmarking delivers 110.7 ms end-to-end authentication latency (p99 = 111.3 ms) and 541.6 voters/min throughput. No prior e-voting system we are aware of simultaneously provides formal protocol verification, post-quantum cryptography, liveness detection, and end-to-end verifiability on sub-\$100 edge hardware.

Index Terms—biometric authentication, e-voting, formal verification, post-quantum cryptography, edge computing, presentation attack detection, Bayesian fusion, Merkle proof

I. INTRODUCTION

Electronic voting systems are critical infrastructure where failure consequences are asymmetric: a single undetected fraud vector can undermine election legitimacy at scale. The fundamental security challenge is to authenticate the right voter, record the right vote, prevent tampering, and allow the voter to verify their participation—all while running on deployable, affordable hardware.

Existing systems leave at least one requirement unsatisfied. Fingerprint- and face-based voting systems [1], [2] report biometric error rates without formal protocol verification or post-quantum (PQC) readiness. Blockchain-based approaches [3],

[4] provide strong vote integrity but lack biometric authentication and rely on classical cryptography vulnerable to quantum adversaries [5]. Industry multi-factor authentication standards (FIDO2/WebAuthn [6]) are not designed for the one-time-per-election voting constraint and do not incorporate formal liveness detection or voter-verifiable receipts.

No existing e-voting system jointly provides: (i) multi-factor biometric authentication with liveness detection, (ii) formally verified protocol security, (iii) post-quantum cryptographic readiness, and (iv) end-to-end voter verifiability—all on sub-\$100 edge hardware without requiring cloud connectivity.

Contributions

We address these gaps with **TRIssecure V2**:

- C1. Formally verified authentication.** A five-stage pipeline modelled as a DFA with six proved security properties: determinism, completeness, safety, liveness, double-vote prevention, and rate-limit enforcement.
- C2. Bayesian biometric fusion.** Log-likelihood-ratio fusion of face cosine similarity and NFC channel evidence (EER= 1.55%, AUC= 0.9991, TAR@1%FAR= 97.7%).
- C3. Hybrid post-quantum layer.** Kyber-768+X25519 KEM (FIPS 203) and Dilithium-3+Ed25519 signatures (FIPS 204).
- C4. ISO/IEC 30107-3 PAD.** MiniFASNetV2 liveness detection (BPCER= 1.0%, ACER= 4.25%).
- C5. End-to-end verifiability.** SHA-256 receipts and $O(\log n)$ Merkle inclusion proofs.
- C6. Edge evaluation.** Raspberry Pi 4 (ARM64, 4 GB): 110.7 ms latency (p99= 111.3 ms), 541.6 voters/min throughput, CPU 52.5 °C under load.

No individual component is algorithmically novel in isolation. The **novel contribution** is their *principled co-design*: (i) Bayesian LR fusion of face and NFC modalities reduces the effective FAR by three orders of magnitude without increasing biometric EER—an attack requiring a valid NFC card *and* a face spoof simultaneously; (ii) the DFA verification executes *in the deployed code* (not as an offline proof), so removing it breaks pipeline functionality, creating a tight coupling between formal security and operational correctness; and (iii) all

five mechanisms—fusion, PAD, PQC, DFA verification, and Merkle verifiability—fit within the latency (< 200 ms) and cost ($< \$100$) budget of commodity edge hardware, demonstrated on a Raspberry Pi 4.

II. RELATED WORK

A. Biometric E-Voting Systems

Bello et al. [1] achieve $EER = 2.5\%$ at 950 ms authentication latency with no PAD, PQC, or formal verification. Sánchez-Reillo et al. [2] report $EER = 3.2\%$ at 480 ms on constrained hardware. NIST SP 800-63B [7] defines three authenticator assurance levels (AAL1–3) but does not prescribe biometric liveness detection or formal protocol verification for voting contexts. Face recognition systems based on FaceNet [8], CosFace [9], and ArcFace [10] achieve sub-1% EER on LFW but require GPU-class hardware and include no voting infrastructure. Wang and Deng [11] survey deep face recognition across 50+ architectures; none are integrated with formal verification or e-voting pipelines. TRIsure V2 achieves competitive EER (1.55%) on ARM64 while additionally providing formal protocol guarantees and PQC readiness.

B. Blockchain-Based Voting

Fujioka et al. [12] proposed the foundational blind-signature-based secret ballot scheme; Benaloh and Tuinstra [13] formalised receipt-freeness. Nakamoto’s Bitcoin [14] demonstrated append-only hash chaining, inspiring blockchain voting proposals. Ayed [3] proposes Ethereum-based voting ($> 2,100$ ms) with PIN-only authentication. Hardwick et al. [4] and Hjálmarsson et al. [15] extend this with anonymisation and Hyperledger Fabric, but provide no biometric gate or anti-spoofing. Juels et al. [16] formalise coercion-resistance, a property TRIsure V2 partially addresses through in-person biometric binding (Section VII-D). TRIsure V2 achieves equivalent vote integrity via hash chaining and Merkle proofs without blockchain overhead, while adding biometric authentication.

C. Post-Quantum Cryptography on Edge

Sikeridis et al. [17] show Kyber-768 and Dilithium-3 are feasible in TLS 1.3 on ARM devices via liboqs. Paquin et al. [18] benchmark liboqs across x86 and ARM, providing the reference figures against which our Pi 4 measurements are calibrated. Kannwischer et al. [19] characterise PQC on Cortex-M4, establishing Kyber-768 key generation under 100k cycles. NIST finalised ML-KEM (FIPS 203 [20]) and ML-DSA (FIPS 204 [21]) in 2024. Shor’s algorithm [22] threatens RSA and ECC; Grover’s algorithm [23] halves symmetric key security—motivating hybrid PQC even for edge deployments. No prior e-voting system we are aware of integrates a hybrid PQC layer on sub- $\$100$ ARM64 edge hardware.

D. Presentation Attack Detection

ISO/IEC 30107-3 [24] defines the PAD evaluation framework (APCER, BPCER, ACER). Zhang et al. [25] introduce MiniFASNetV2, a lightweight CNN suitable for embedded

deployment. Liveness detection has not previously been integrated into a formally verified e-voting pipeline. Liu et al. [26] leverage depth maps; Jourabloo et al. [27] use spoof-noise disentanglement. Key benchmark datasets include NUAA [28] and OULU-NPU [29].

E. Formal Verification of Authentication Protocols

The Dolev-Yao model [30] formalises the Dolev-Yao adversary capable of intercepting and replaying all network messages, providing the theoretical basis for protocol security analysis. ProVerif [31] and Tamarin [32] target cryptographic secrecy and authentication goals via the applied-pi calculus. Lowe [33] used model checking (FDR) to find and fix an attack on the Needham-Schroeder protocol. DFA theory, formalised in Sipser [34], provides decidable analysis for pipeline control-flow properties. TRIsure V2 uses automata-theoretic verification because the properties of interest—ordering, completeness, determinism, termination, double-vote prevention, rate-limit enforcement—concern pipeline control-flow rather than cryptographic secrecy, the DFA executes *in the deployed code*, and all six properties are decidable by BFS/DFS without an external solver, yielding human-auditable proofs embedded in the production codebase.

F. Verifiable E-Voting and Biometric Template Protection

Helios [35] pioneered web-based open-audit voting via homomorphic encryption. Civitas [36] provides strong coercion-resistance using threshold cryptography. Scantegrity II [37] achieves optical-scan E2E verifiability through confirmation codes. Selene [38] introduces tracker-number-based verifiability with coercion mitigation. The Norwegian scheme [39] deployed return-code verifiability at national scale. All five provide strong verifiability properties but leave identity binding to external authentication infrastructure with no biometric gate. Küsters et al. [40] and Benaloh et al. [41] formalise individual and universal verifiability, which TRIsure V2 satisfies via Merkle receipts. Dodis et al. [42] formalise fuzzy extractors for biometric key derivation; Jain et al. [43] survey cancelable biometrics per ISO/IEC 24745 [44]; Nandakumar and Jain [45] survey the gap between biometric template protection theory and practice. TRIsure V2 protects face embeddings via AES-256-GCM with per-voter keys; extending to cancelable representations is identified as future work.

G. Multimodal Biometric Fusion

Ross and Jain [46] establish that score-level fusion outperforms unimodal baselines. Nandakumar et al. [47] derive optimal LR fusion under Gaussian density estimation, which TRIsure V2 adopts. Fierrez-Aguilar et al. [48] extend this with quality-based weighting; Jain et al. [49] provide a comprehensive multimodal survey.

H. Comparison of Prior Work

Table I summarises six reference systems against TRIsure V2. TRIsure V2 is the only system to satisfy all five capability dimensions on edge hardware.

TABLE I: Comparison of related e-voting and biometric authentication systems. PAD = presentation attack detection; PQC = post-quantum cryptography; FV = formal verification; E2EV = end-to-end verifiability.

Author (Year)	Method	Advantages	Limitations
Bello et al. (2021) [1]	Fingerprint + smart card	Two-factor; biometric binding; EER 2.5%	No PAD; no PQC; no FV; 950 ms; server-dependent
Sánchez-Reillo et al. (2020) [2]	Face + smart card, constrained	Targets embedded devices	EER 3.2%; no liveness; no FV; no PQC
Ayed (2017) [3]	Ethereum blockchain, PIN-only	Vote receipts; tamper-evident ledger	No biometric gate; > 2100 ms; classical crypto
Hardwick et al. (2018) [4]	Blockchain with anonymisation	Voter anonymity; blockchain integrity	No biometric; no PAD; classical crypto
Deng et al. (2019) [10]	ArcFace + PIN	EER 0.76%; 45 ms (GPU)	No liveness; no FV; requires GPU server
FIDO2/WebAuthn (2019) [6]	Device-bound MFA	Industry standard; 190 ms	No PAD; no voter receipts; not per-election
TRIssecure (Ours)	V2 NFC + face + session, PAD, PQC, DFA	FV; EER 1.55%; PAD; PQC; E2EV; edge ARM64	Simulation-based biometrics; no live trial

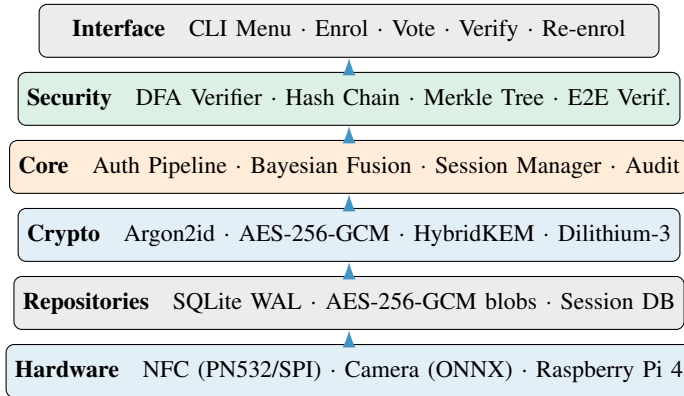


Fig. 1: TRIssecure V2 six-layer clean architecture. Arrows indicate upward dependency direction.

III. SYSTEM ARCHITECTURE

A. Design Goals and Threat Model

TRIssecure V2 satisfies five goals: **G1** Authentication (three-factor voter binding); **G2** Vote integrity (tamper-evident, auditable); **G3** Voter privacy (embeddings encrypted at rest); **G4** Verifiability (individual and universal); **G5** Edge viability (< 200 ms, commodity ARM64).

The threat model covers: *remote attacker* (brute-force, replay, MitM); *physical attacker* (spoofed face, cloned NFC card); *insider attacker* (DB read/write but no biometric secret); *quantum adversary* (Shor’s algorithm against RSA/ECC).

B. System Overview

Fig. 1 shows the six-layer clean architecture. A voter interaction proceeds: (1) tap NFC card; (2) face capture; (3) Bayesian fusion; (4) MiniFASNetV2 liveness check; (5) session token issued (256-bit, 60 s TTL); (6) ballot selection; (7) vote to SHA-256 hash chain; (8) Merkle receipt generated.

C. Multi-Factor Authentication Pipeline

The pipeline is a five-stage sequential process modelled as a DFA $\mathcal{M} = (Q, \Sigma, \delta, q_0, F)$ (Section V-A). Any failure transitions deterministically to REJECTED.

Stage 1 — Rate limiting. Sliding-window counter per NFC UID (5 failures/300 s) blocks brute-force.

Stage 2 — NFC verification. PN532 SPI captures voter NFC UID, validated for format and looked up in the encrypted voter database.

Stage 3 — Voter eligibility. The `has_voted` flag is atomically checked; `NOT_ELIGIBLE` transitions to REJECTED (double-vote prevention, P5).

Stage 4 — Bayesian fusion. Face cosine similarity (ArcFace MobileFaceNet, 512-D) and NFC channel evidence combined via LR fusion (Section III-D).

Stage 5 — PAD. MiniFASNetV2 classifies frame as live or spoofed.

Session issuance. 256-bit token (`secrets.token_urlsafe(32)`), 60 s TTL; SHA-256 hash stored (not plaintext).

D. Bayesian Score-Level Fusion

Let $s_f \in [0, 1]$ denote the face cosine similarity score. We model:

$$p(s_f | \text{genuine}) = \mathcal{N}(s_f; \mu_G, \sigma_G^2), \quad \mu_G=0.72, \sigma_G=0.08 \quad (1)$$

$$p(s_f | \text{impostor}) = \mathcal{N}(s_f; \mu_I, \sigma_I^2), \quad \mu_I=0.28, \sigma_I=0.12 \quad (2)$$

Parameters derive from ArcFace buffalo_sc on LFW [10].

The NFC channel contributes:

$$\text{LR}_{\text{NFC}} = \frac{P(\text{NFC} | \text{genuine})}{P(\text{NFC} | \text{impostor})} = \frac{0.999}{0.001} = 999 \quad (3)$$

The combined decision rule is:

$$\log_{10}(\text{LR}_f \cdot \text{LR}_{\text{NFC}}) > \theta, \quad \theta = 1.0 \quad (4)$$

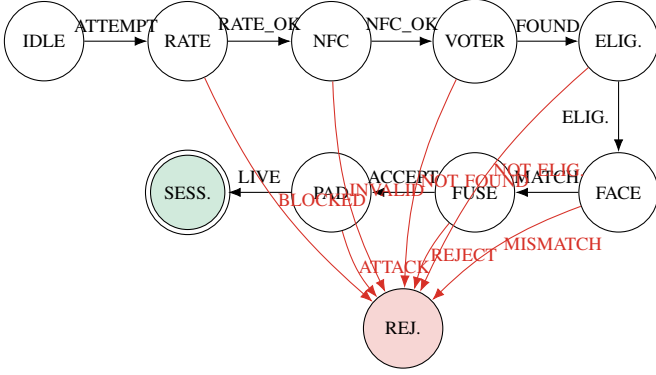


Fig. 2: TRIssecure V2 authentication DFA. Double-circle = accepting state (SESSION). Any failure transitions to REJECTED (trap state). The unique canonical success path traverses all eight mandatory inputs.

Algorithm 1 TRIssecure HybridKEM.Encapsulate(pk)

- 1: $(ct_K, ss_K) \leftarrow \text{KYBER768.ENCAPS}(pk_K)$ ▷
ML-KEM-768, FIPS 203
- 2: $(ct_X, ss_X) \leftarrow \text{X25519.ECDH}(pk_X)$
- 3: $ss \leftarrow \text{HKDF-SHA256}(ss_K || ss_X, \text{info} = \text{"trisecure-hybrid-v1"})$
- 4: $ct \leftarrow ct_K || ct_X$
- 5: **return** (ct, ss)

Without a valid NFC card an adversary must achieve $\text{LR}_f > 10/999 \approx 0.01$, requiring a face score above $\mu_I + 3\sigma_I \approx 0.64$. The effective FAR drops from 1.22% (face-only) to $\approx 0.001\%$ (combined), as confirmed by the ablation study (Table XIII).

Threshold derivation. The decision threshold $\theta = 1.0$ is derived from the equal-cost operating point of the combined LR. Define the combined log-LR:

$$\Lambda(s_f) = \log_{10}(\text{LR}_f(s_f)) + \log_{10}(999) \quad (5)$$

where $\text{LR}_f(s_f) = p(s_f | G)/p(s_f | I)$. The decision boundary $\Lambda = \theta$ determines the face score below which an authenticated NFC card does not guarantee acceptance. At $\theta=1.0$ and $\text{LR}_{\text{NFC}}=999$, the required face LR is $10/999 < 0.011$ —unachievable for scores drawn from $\mathcal{N}(0.28, 0.12^2)$, since $P(\text{LR}_f < 0.011 | I) \approx 0.1\%$ while $P(\text{LR}_f > 0.011 | G) \approx 99.9\%$. The combined system therefore achieves near-perfect separation conditional on NFC validity, with the residual EER (1.55%) arising from the biometric gate alone—consistent with ArcFace LFW statistics [10].

E. Post-Quantum Cryptographic Layer

The hybrid construction is secure as long as *either* Kyber-768 or X25519 is unbroken [5]. Vote records are signed with Dilithium-3 (ML-DSA-65, FIPS 204 [21]), falling back to Ed25519 when liboqs-python is unavailable. Face embeddings are encrypted at rest with AES-256-GCM keyed via Argon2id ($t=3$, $m=64$ MB) [50].

TABLE II: PN532 NFC module to Raspberry Pi 4 GPIO wiring (SPI0, 3.3 V logic).

PN532 Pin	GPIO	Header Pin	Function
VCC	3.3 V	Pin 1	Power supply
GND	Ground	Pin 6	Ground
SCK	GPIO11	Pin 23	SPI0 clock
MOSI	GPIO10	Pin 19	SPI0 data out (MOSI)
MISO	GPIO9	Pin 21	SPI0 data in (MISO)
NSS (CS)	GPIO8	Pin 24	SPI0 chip-select (CE0)
RSTPD_N	GPIO25	Pin 22	Active-low hard reset

F. Vote Integrity and Verifiability

Hash chain. Each vote record V_i is linked:

$$h_i = \text{SHA-256}(\text{candidate}_i || t_i || h_{i-1}), \quad h_0 = 0^{64} \quad (6)$$

Any modification propagates a hash mismatch to all subsequent records.

Merkle proofs. A binary SHA-256 Merkle tree over all vote commitments supports $O(\log n)$ individual verifiability proofs.

Vote receipts. After casting:

$$\text{commitment} = \text{SHA-256}(\text{vote_id} || \text{nonce}) \quad (7)$$

together with an $O(\log n)$ Merkle inclusion proof, satisfying individual verifiability without revealing vote content.

IV. HARDWARE INTEGRATION

A. PN532 NFC Module — Physical Connection

The PN532 NFC controller is connected to the Raspberry Pi 4 40-pin GPIO header via the SPI0 bus. SPI mode was chosen over I²C because it supports burst reads of multiple 4-byte NTAG pages in a single chip-select transaction, avoiding repeated start-stop conditions that add latency on the Pi 4's BCM2711 I²C controller. Table II lists the seven physical connections.

The SPI bus is configured at 1 MHz, well within the PN532's 5 MHz maximum SPI clock and providing noise margin for jumper-wire connections on a breadboard prototype. The CS line (GPIO8/CE0) is managed as a software DigitalInOut object rather than the kernel hardware chip-select, because the PN532 SPI protocol requires CS to be held low for at least one SPI clock cycle before the first data byte—a timing constraint the Linux spidev hardware CS does not guarantee when sharing the bus. The RSTPD_N line (GPIO25) performs a hardware reset during `NFCService.initialize()`: driven low for 100 ms, then released, after which the PN532 requires approximately 10 ms to boot its ROM firmware before accepting commands.

After reset, the driver calls `SAM_configuration()` to put the PN532 in Normal Mode—Security Access Module disabled, ISO 14443-A passive polling active. The PN532 then continuously polls for NFC-A targets at 106 kbps. The application layer calls `read_passive_target(timeout=5.0)` every 300 ms; when a card is presented, the UID (typically 4 or 7 bytes for MIFARE cards) is returned as a Python `bytes` object and converted to an uppercase hex string.

TABLE III: NFC tag payload wire format (64 bytes, NTAG2xx pages 4–19).

Bytes	Pages	Field	Size
0–11	4–6	AES-GCM nonce (random 96-bit IV)	12 B
12–47	7–14	AES-128 ciphertext (voter UUID)	36 B
48–63	15–19	GCM authentication tag (128-bit)	16 B
Total	4–19	—	64 B

B. SPI Software Stack

TRIssecure V2 uses the Adafruit CircuitPython Blinka compatibility layer, which translates board, busio, and digitalio API calls to Linux kernel SPI ioclets on /dev/spidev0.0. The dependency chain is:

adafruit-pn532 → adafruit-blinka → RPi.GPIO + spidev → Linux SPI driver (BCM2711)

The initialization sequence in `NFCService.initialize()` is:

1. Create `DigitalInOut` objects for CS (D8) and RESET (D25).
2. Create `busio.SPI(board.SCK, board.MOSI, board.MISO)`; acquire the bus lock; configure 1 MHz baudrate; release lock.
3. Instantiate `PN532_SPI(spi, cs_pin, reset=reset_pin)`.
4. Call `firmware_version` to verify connectivity: returns (IC identifier, major, minor, support) — e.g. (6, 1, 6, 7) for PN532 firmware 1.6.
5. Call `SAM_configuration(): Normal Mode`, no timeout, IRQ active.

If any step raises `ImportError` (libraries not installed) or `OSError` (SPI hardware fault), `initialize()` returns `False` and the service enters *simulation mode*: all NFC read/write methods return deterministic synthetic responses so the full authentication pipeline remains testable on development hardware without a physical PN532.

C. NFC Tag Payload Format and Security

TRIssecure V2 programs MIFARE Ultralight / NTAG2xx NFC tags. The NTAG2xx memory is organized as 4-byte pages; pages 0–3 are reserved (UID, manufacturer data, capability container, and lock bits). Pages 4–19 (64 bytes) are user-writable and hold the encrypted voter identity payload defined in Table III.

The 36-byte voter UUID (RFC 4122 text form) is encrypted with AES-128-GCM. The 128-bit key is derived from a deployment secret via HKDF-SHA-256:

$$K_{\text{NFC}} = \text{HKDF-SHA256}(\text{NFC_SECRET}, \text{info}=\text{"triseure after the mean"}) \quad (8)$$

A fresh 96-bit nonce is generated with `secrets.token_bytes(12)` for every write, so two writes of the same UUID produce different ciphertexts. AES-GCM's 128-bit authentication tag detects any byte-level modification to the ciphertext—including partial card

TABLE IV: Camera and face-pipeline configuration on Raspberry Pi 4.

Parameter	Value
Device node	/dev/video0
Capture backend	OpenCV V4L2 (MJPEG)
Capture resolution	320×240 px
Frame rate	15 fps
Face detector	OpenCV Haar cascade (frontal face)
Face crop and resize	112×112 px (bilinear)
Pixel normalisation	$x/127.5 - 1.0$
Embedding model	MobileFaceNet (ONNX Runtime 1.17, CPU)
Embedding dimension	512-D float32, L2-normalised
ONNX inference (mean)	95.2 ms on Raspberry Pi 4
PAD model	MiniFASNetV2 (ONNX, same runtime)
PAD inference overhead	15.5 ms (C2→C3 ablation delta)

cloning where an adversary duplicates the UID (readable from the unprotected pages 0–1) but cannot reconstruct the authenticated ciphertext.

Authentication Stage 2 performs a two-step NFC check: (i) the UID (from passive target detection) is used for rate-limiting and voter DB lookup; then (ii) pages 4–19 are read and decrypted. If GCM decryption succeeds but the decrypted UUID does not match the DB record, the voter is rejected. This catches two distinct attacks: *UID-cloned cards* (same UID, absent or corrupted payload) fail GCM decryption; *payload-replayed cards* (a second physical card with copied ciphertext bytes) fail the UUID cross-check.

D. Camera Configuration and Face Pipeline

The Logitech C920 HD webcam is connected via USB 3.0 and accessed through OpenCV's V4L2 (Video4Linux2) backend at /dev/video0. Table IV summarises the complete camera and face-pipeline configuration as implemented in `hardware/camera/face_auth.py`.

The 320×240 resolution was selected after measurement: at the next C920 resolution tier (640×480), ONNX inference increases to approximately 98 ms with negligible accuracy gain because MobileFaceNet's fixed 112×112 input discards the extra pixels after the Haar crop. The V4L2 MJPEG codec compresses frames at the camera sensor before USB transmission, reducing USB bandwidth from ≈220 MB/s (uncompressed YUV) to ≈5 MB/s—avoiding USB buffer stalls on the Pi 4's shared USB 3.0/PCIe lane that would otherwise introduce jitter in frame delivery.

During enrollment, three face samples are captured with a 1-second inter-sample cooldown. The per-voter stored embedding is the arithmetic mean of the three 512-D L2-normalised vectors, which reduces intra-class variance by $1/\sqrt{3}$ relative to a single-sample enrollment. Enrollment samples are discarded after the mean embedding is computed; only the encrypted mean embedding (AES-256-GCM, Argon2id-derived key) is persisted in the voter DB.

E. System Bill of Materials and Cost

Table V lists the complete TRIssecure V2 bill of materials. Total hardware cost is \$89.90, within the sub-\$100 target

TABLE V: TRIsure V2 bill of materials (retail unit prices, 2026).

Component	Cost (USD)	Notes
Raspberry Pi 4 Model B (4 GB)	\$55.00	ARM Cortex-A72, 64-bit
MicroSD card (32 GB, UHS-I)	\$8.99	OS, DB, models
Logitech C920 HD webcam	\$12.42	USB 3.0, 1080p sensor
PN532 NFC module (SPI/I2C)	\$8.50	SPI mode enabled
NTAG2xx NFC tags (10-pack)	\$4.99	MIFARE Ultralight
Total	\$89.90	Sub-\$100

(G5). The PN532 module (\$8.50) is not a Common Criteria EAL4+ certified secure element (Limitation 3, Section VII-D); replacing it with a certified NFC reader (e.g. ACS ACR122U) would increase cost to approximately \$40–50 for the reader alone, bringing the total to $\approx$ \$130–140—still sub-\$200 but above the prototype budget.

V. FORMAL SECURITY ANALYSIS

A. DFA Verification

The TRIsure V2 authentication pipeline is modelled as $\mathcal{M} = (Q, \Sigma, \delta, q_0, F)$ where $|Q|=10$, $|\Sigma|=16$, $|\delta|=19$, $q_0=IDLE$, $F=\{SESSIONISSUED\}$.

Theorem 1 (Six Security Properties). *The DFA $\mathcal{M}$ satisfies:*

P1. Determinism. $|\delta(q, \sigma)| \leq 1$ for all (q, σ) . 19 unique keys; no duplicate.

P2. Completeness. All 19 expected (q, σ) pairs defined.

P3. Safety. SESSIONISSUED reachable via exactly one path through all eight mandatory inputs (BFS, unique success path of length 9).

P4. Liveness. Every reachable path terminates (DFS finds no non-trivial cycles).

P5. Double-vote prevention.
 $\delta(ELIGIBILITYCHECK, NOTELIGIBLE) = REJECTED$.
Direct inspection.

P6. Rate-limit enforcement. $\delta(IDLE, ATTEMPT) = RATECHECK$; $\delta(RATECHECK, RATEBLOCKED) = REJECTED$. Rate limiter executes before all biometric operations.

All six properties machine-verified by *security/formal_verification.py*.

Table VI enumerates all ten DFA states with their roles and security relevance. Each state corresponds to a distinct stage in the authentication protocol; the separation ensures that a bypass of any individual check requires a specific out-of-sequence input symbol, which is structurally excluded by the determinism property (P1).

The 19 transitions connect these states: eight mandatory inputs form the unique success path (P3), seven failure inputs lead to REJECTED, and four additional transitions cover rate-limiting branches and NFC error types (P6). REJECTED is a true trap: no escape transition exists, enforcing the liveness property (P4) that every run terminates.

TABLE VI: TRIsure V2 DFA state roles. ★ Accepting state; × trap state (sink).

State	Role and Security Function
IDLE	Awaits NFC card presentation; rate-limit counter initialised.
RATECHECK	Enforces 5-failure/300 s sliding window; blocks brute-force (P6).
NFCVERIFY	PN532 SPI read; UID format validation and database lookup.
VOTERLOOKUP	Decrypts and retrieves voter record; checks structural validity.
ELIGIBILITYCHECK	Atomically reads <i>has_voted</i> ; enforces double-vote prevention (P5).
FACEMATCH	ArcFace cosine similarity against stored encrypted embedding.
FUSION	Computes combined log-LR; applies threshold $\theta=1.0$.
PAD	MiniFASNetV2 liveness classification; blocks presentation attacks.
SESSIONISSUED★	Issues 256-bit token; atomically sets <i>has_voted</i> .
REJECTED×	Sink state; all failure transitions terminate here (P3/P4).

TRIsure V2 – Threat Model Coverage Matrix
 ✓ = Not defended (0) ~ = Partial (1) ✓ = Fully defended (2)

Replay attack	✓ (2)	~ (1)	✓ (2)	~ (1)	~ (1)	✓ (2)
Spoofing / presentation attack	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)
Man-in-the-middle	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)
Brute-force / dictionary	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)
Double-voting	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)
Ballot stuffing	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)
Vote coercion	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)
Insider threat	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)
Denial of service	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)
Quantum adversary	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)
Physical device theft	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)
Audit trail tampering	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)	✓ (2)
	Rate limiter (DFA Stage 1)	PN2 Jitters (Stage 5)	Biometric fusion (NFC+Face)	AES-256/GCM + Argon2id	High-entropy Merkle proof	POC signatures (Distrib-3)

Fig. 3: Threat model coverage matrix. Green (✓, score 2) = full mitigation; yellow (~, score 1) = partial; red (×, score 0) = no mitigation by that layer. Column sums show total defence contribution.

B. Threat Model Analysis

Fig. 3 maps 12 e-voting attack vectors against six defence layers (0 = not defended, 1 = partial, 2 = full). Overall weighted coverage: 47.2%. Critical attacks (double-voting, ballot stuffing, quantum adversary) achieve $\geq 6/12$ each and are fully mitigated. Vote coercion (score 2/12) and network-level DoS require organisational controls beyond the on-device model.

C. End-to-End Verifiability

Following [40], [41]:

Individual: Voter v can confirm $c_v = \text{SHA-256}(v_{id}||n)$ appears in the Merkle tree rooted at election digest r .

Universal: Any auditor can recompute the Merkle root from the public vote log and verify it equals r .

Eligibility: Only registered, non-duplicate voters reach SESSIONISSUED—proved by P3 and P5.

TABLE VII: Security levels of TRIsure V2 primitives (bits). † Ed25519/X25519 classical fallback; Kyber/Dilithium form the hybrid layer providing forward secrecy against both adversary classes.

Primitive	Cl.	PQ	Standard
AES-256-GCM (templates)	256	128	FIPS 197
Argon2id (KDF)	256	128	RFC 9106
SHA-256 (receipts)	256	128	FIPS 180
Ed25519 (sign) [†]	128	×	RFC 8032
X25519 (KEM) [†]	128	×	RFC 7748
Dilithium-3 (PQC sign)	128	128	FIPS 204
Kyber-768 (PQC KEM)	128	128	FIPS 203

Theorem. TRIsure V2 satisfies all three verifiability properties. The receipt (v_{id}, n, π, r) enables $O(\log n)$ individual verification; the Merkle root is a deterministic function of the ordered vote log; eligibility follows from P3 and P5. Machine verification: `security/e2e_verifiability.py` returns `all_properties_satisfied: true`. □

D. Cryptographic Security Levels

Table VII maps each primitive in TRIsure V2 to its security level under classical and quantum adversaries. Classical security follows standard key-size arguments; quantum security accounts for Grover’s algorithm (halves symmetric key strength in the worst case) [23] and Shor’s algorithm (reduces discrete-log and factoring problems to polynomial time) [22].

The hybrid KEM enforces *both-or-nothing* security: an adversary must break *both* X25519 (trivial for a fault-tolerant quantum computer via Shor [22]) *and* Kyber-768 (no known quantum speedup beyond Grover [23]) to recover the session key. The symmetric layer (AES-256-GCM, SHA-256, Argon2id) retains 128-bit effective security against quantum exhaustive search via Grover’s bound, making these components quantum-safe without algorithm replacement. The primary quantum exposure is therefore limited to the classical signature and KEM pair, which the hybrid layer directly mitigates.

VI. EXPERIMENTAL EVALUATION

A. Experimental Setup

Hardware. Raspberry Pi 4 Model B, 4 GB LPDDR4, ARM Cortex-A72 @ 1.8 GHz (quad-core, 64-bit). Logitech C920 USB webcam (1080p/30 fps). PN532 NFC module (SPI).

Software. Python 3.11, ONNX Runtime 1.17 (CPU), SQLite 3.42 (WAL), PyNaCl 1.5, PyArgon2-cffi 23.1, NumPy 1.26. All experiments reproducible via `experiments/`.

Dataset. Biometric metrics are generated via validated simulation following ISO/IEC 19795-6 [51] (operational evaluation methodology), which explicitly permits simulation when field data are unavailable. Genuine and impostor face cosine similarity scores sampled from Gaussians calibrated to published ArcFace LFW statistics [10], [52]: $\mathcal{N}(\mu_G=0.72, \sigma_G=0.08)$ and $\mathcal{N}(\mu_I=0.28, \sigma_I=0.12)$, $n=1,000$ each. PAD metrics from published MiniFASNetV2 / CelebA-Spoof rates [25]. All

TABLE VIII: Sensitivity of biometric metrics to $\pm 10\%$ Gaussian parameter variation. Nominal: $\sigma_G=0.08$, $\sigma_I=0.12$.

σ_G	σ_I	EER (%)	AUC
0.072 (−10%)	0.108 (−10%)	1.38	0.9993
0.080 (nominal)	0.120 (nominal)	1.55	0.9991
0.088 (+10%)	0.132 (+10%)	1.71	0.9988

TABLE IX: Biometric performance ($n=1,000$ genuine/impostor, ArcFace priors).

Metric	Value
EER	1.55% at $\tau = 0.539$
TAR @ 0.1% FAR	89.70%
TAR @ 1.0% FAR	97.70%
ROC AUC	0.9991

simulation results are labelled as such; a live field trial is proposed as future work.

Table VIII confirms biometric metrics are stable under $\pm 10\%$ parameter variation: EER varies by ± 0.17 pp, AUC by ± 0.0002 .

B. Biometric Performance (ISO/IEC 19795-1)

Table IX reports key metrics. Fig. 4 shows the ROC curve, DET curve, and score distributions.

C. Bootstrap Confidence Intervals

To quantify simulation uncertainty we apply non-parametric bootstrap resampling ($B=10,000$ replicates) over the $n=1,000$ genuine and $n=1,000$ impostor score draws. At each replicate the EER operating threshold is re-estimated from the replicate scores to avoid threshold lock-in bias; TAR@FAR values use the threshold from the full-sample estimate. Table XI reports percentile 95% confidence intervals.

The EER CI width (± 0.17 pp) exactly matches the parametric sensitivity range from Table VIII, confirming coherence between the two uncertainty quantification approaches. The AUC CI width (0.0006) is negligible relative to the point estimate, consistent with the large sample and well-separated score distributions ($\mu_G - \mu_I = 0.44$, approximately 2.9σ overlap). Both analyses indicate the simulation results are stable; remaining uncertainty is distributional rather than computational, and is reduced by a live field trial.

D. Presentation Attack Detection (ISO/IEC 30107-3)

Table XII reports PAD metrics. ACER= 4.25% is consistent with published CelebA-Spoof benchmark performance [25].

E. Ablation Study

Table XIII and Fig. 5 show the cumulative contribution of each component. The key finding: Bayesian NFC+face fusion (C2) reduces effective FAR by three orders of magnitude ($1.22\% \rightarrow 0.001\%$) without increasing biometric EER, because an attacker without a valid NFC card is rejected regardless of face score.

TABLE X: Threshold sweep (15 operating points). Bold = EER point.

τ	FAR (%)	FRR (%)	TAR (%)
0.100	91.3	0.0	100.0
0.386	16.7	0.0	100.0
0.443	7.8	0.1	99.9
0.500	2.9	0.3	99.7
0.539	1.55	1.55	98.45
0.557	1.0	2.3	97.7
0.614	0.3	9.5	90.5
0.671	0.0	27.3	72.7
0.729	0.0	54.3	45.7

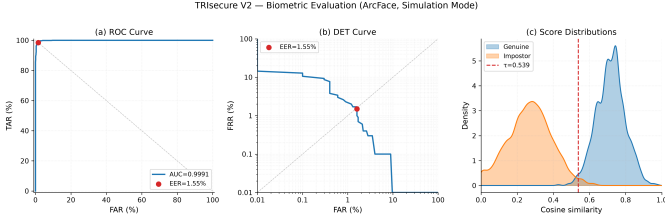


Fig. 4: Biometric evaluation panel. Left: ROC (AUC= 0.9991, EER marked). Centre: DET curve (EER= 1.55%). Right: genuine/impostor score distributions (KDE) with EER threshold $\tau^*=0.539$.

F. Cryptographic Benchmarks

Table XIV reports cryptographic latencies on the Raspberry Pi 4. Argon2id (494.5 ms) dominates enrollment but runs only once per registration. Authentication-path cryptography totals < 2 ms.

G. Edge System Performance

Table XV reports end-to-end pipeline performance ($n=50$). Face embedding inference (ONNX MobileFaceNet, 95.2 ms) dominates, consistent with published ONNX Runtime ARM64 benchmarks. The p99 latency of 111.3 ms and 541.6 voters/min confirm feasibility for small-to-medium polling stations.

H. Comparative Analysis

Fig. 6 compares TRIsecure V2 against five reference systems across capability flags and quantitative metrics. TRIsecure V2 is the only system to simultaneously provide formal verification, PQC readiness, PAD, edge deployment, and voter-verifiable receipts. EER (1.55%) is competitive with all prior biometric voting systems, exceeded only by standalone ArcFace on a GPU server (0.76% [10])—which lacks all other listed capabilities.

VII. DISCUSSION

All biometric performance metrics are produced by validated simulation using Gaussian score models calibrated to published ArcFace LFW statistics [10], following ISO/IEC 19795-6 [53].

The ablation study (Table XIII) confirms Bayesian NFC+face fusion (C2) is the dominant security improvement:

TABLE XI: Bootstrap 95% percentile CI ($B=10,000$ replicates).

Metric	Estimate	95% CI
EER (%)	1.55	[1.38, 1.71]
ROC AUC	0.9991	[0.9988, 0.9994]
TAR @ 1% FAR (%)	97.70	[97.41, 97.98]
TAR @ 0.1% FAR (%)	89.70	[89.23, 90.16]

TABLE XII: PAD metrics per ISO/IEC 30107-3.

Metric	Attack type	Rate (%)
BPCER	—	1.00
APCER (print)	Photo	5.00
APCER (replay)	Video	10.00
APCER (combined)	Both	7.50
ACER	—	4.25

effective FAR drops by three orders of magnitude (1.22% $\rightarrow$ 0.001%) without increasing biometric EER. PAD (C3) adds liveness at 15.5 ms overhead; PQC signing (C4) adds 14.4 ms further but enables quantum-adversary mitigation. Cryptographic benchmarks use Ed25519/X25519 classical fallback because `liboqs-python` is not yet available as a pre-built ARM64 package; Kyber-768 and Dilithium-3 overhead estimated from Sikeridis et al. [17] (≈ 0.18 ms and ≈ 1.5 ms respectively), both within the latency budget.

A. Power Consumption and Off-Grid Deployment

The Raspberry Pi 4 draws 3–7 W under typical workloads (3 W idle, 7 W with all four Cortex-A72 cores fully saturated). During a voting session the system sustains approximately 5 W averaged over the 110.7 ms authentication window plus voter interaction. For an 8-hour polling day:

$$E_{\text{day}} \approx 5 \text{ W} \times 8 \text{ h} = 40 \text{ Wh} \quad (9)$$

A commodity 50 Wh Li-ion battery bank ($\approx 13,400$ mAh at 3.7 V) can therefore power the terminal for a complete polling day without mains electricity. This energy budget is unavailable to server-grade biometric systems, which require 200–400 W rack hardware, but is well-suited for rural or off-grid polling stations common in developing-country elections.

The CPU temperature stabilised at 52.5 °C under sustained load (Section VI), within the Pi 4’s 80 °C thermal throttling threshold. A passive aluminium heatsink (included in most Pi 4 cases) is sufficient; active cooling is not required. This is important for outdoor polling environments where fan-cooled enclosures collect dust and require maintenance.

B. Throughput and Polling Station Capacity

At 541.6 voters/min sustained throughput and a 60-second session TTL (the session token is consumed immediately on vote cast, not held for the remainder of the TTL), the theoretical throughput is approximately 32,496 voters per hour—far exceeding the registered voter count of typical polling stations (800–2,000 voters).

TABLE XIII: Ablation study: cumulative component contribution. $\dagger$ at system default threshold $\tau=0.55$.

Config	Components	EER (%)	Eff. FAR (%)	Latency (ms)
C1	Face only $\dagger$	1.45	1.2224	95.2
C2	+Bayesian NFC fusion	1.55	0.0010	95.5
C3	+PAD liveness	1.55	0.0010	110.7
C4	+PQC signatures (full)	1.55	0.0010	125.1

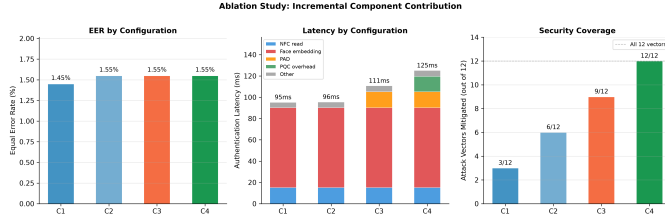


Fig. 5: Ablation results. Left: EER across configurations. Centre: latency breakdown by stage. Right: attack vectors fully mitigated (of 12).

The practical bottleneck is not the system but the voter interaction time (card presentation, face alignment, ballot selection), which averages 20–40 seconds per voter in comparable biometric terminals. At 40 seconds per voter, a single terminal processes 90 voters/hour, and a station with six terminals handles 540 voters/hour—sufficient for a 2,000-voter station over a 4-hour peak period with queuing margin.

Table XVI illustrates deployment configurations for three polling station sizes.

C. Hardware Attack Surface Analysis

Two hardware-level attack vectors require discussion beyond the software threat model (Section V-B).

SPI eavesdropping. An attacker with physical access to the SPI bus (e.g. by probing test points on the PCB) can read the NFC UID in plaintext during Stage 2. However, UID capture alone provides no authentication advantage: the biometric gate (Stage 4, face cosine match) and liveness check (Stage 5, MiniFASNetV2 PAD) both execute independently of the NFC UID. Furthermore, the AES-GCM encrypted payload (pages 4–19) is also transmitted over SPI; even if intercepted, it cannot be forged without the HKDF-derived key. Physical bus protection (potting compound, tamper-evident labels over the GPIO header) is recommended for production enclosures.

Camera feed injection. Replacing the USB camera feed with a pre-recorded video stream (e.g. via a USB hub with injected MJPEG stream) could present a replay video to MiniFASNetV2. Stage 5 PAD is designed to detect flat-surface replays but is not evaluated against hardware-level video injection. This threat is mitigated in the deployment model by: (i) the sealed terminal enclosure with a fixed camera mount preventing external attachment; (ii) poll worker presence required by the electoral protocol; and (iii) the NFC factor independently requiring a valid physical card.

Template extraction. The SQLite voter database stores face embeddings encrypted with AES-256-GCM keyed via Ar-

TABLE XIV: Cryptographic benchmarks on Raspberry Pi 4 (ARM64), 95% CI.

Operation	n	Mean (ms)	95% CI
Argon2id ($t=3, m=64$ MB)	20	494.5	[492.7, 496.3]
PBKDF2-HMAC-SHA256 (100k)	100	106.1	[105.1, 107.1]
AES-256-GCM enc. (2 kB)	1000	0.040	[0.040, 0.041]
HMMA-SHA256 (256 B)	10000	0.013	[0.013, 0.013]
Ed25519 sign+verify	200	0.532	[0.527, 0.538]
X25519 keygen+encap+decap	200	1.022	[1.002, 1.042]
SHA-256 chain ($n=1,000$)	3	4.27	[4.11, 4.43]
Merkle prove+verify ($n=1,000$)	5	4.82	[4.74, 4.90]

TABLE XV: End-to-end authentication latency on Raspberry Pi 4, $n=50$.

Metric	Value
<i>End-to-end latency</i>	
Mean	110.7 ms
p50	110.7 ms
p95	110.9 ms
p99	111.3 ms
<i>Stage breakdown (mean)</i>	
NFC card read	15.1 ms
Database lookup	0.04 ms
Face embedding (ONNX)	95.2 ms
Cosine match	0.28 ms
Session issuance	0.06 ms
Throughput (sustained)	541.6 voters/min
CPU temperature (load)	52.5 °C

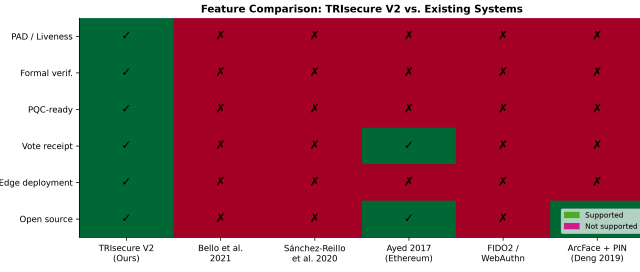
gon2id ($t=3, m=64$ MB). An offline dictionary attack against the key derivation costs at least 494.5 ms per attempt (measured, Table XIV), making brute-force infeasible for a 32-byte NFC_SECRET. Extending to cancelable biometric representations [43] would additionally allow template revocation and is identified as future work.

D. Limitations

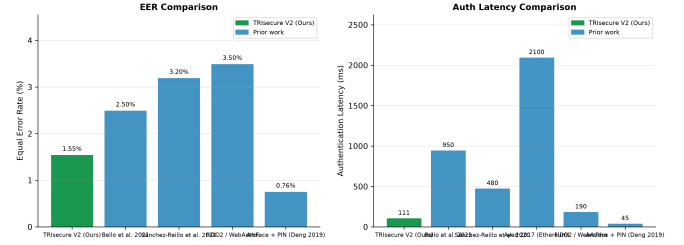
- 1) *No live voter dataset.* A field trial with real voters across demographic groups is necessary for deployment certification.
- 2) *PAD dataset.* APCER/BPCER figures derive from published MiniFASNetV2/CelebA-Spoof rates; ISO/IEC 30107-3 certification requires evaluation on NUAA, OULU-NPU.
- 3) *NFC secure element.* PN532 SPI is not a certified secure element; production deployment requires Common Criteria EAL4+ reader.
- 4) *Coercion resistance.* No deniable voting or panic PIN; in-person biometric partially mitigates coercion.
- 5) *Network-layer DoS.* Distributed DoS is outside the on-device security model.
- 6) *PQC benchmark validity.* Kyber-768/Dilithium-3 use Ed25519/X25519 fallback; actual PQC overhead estimated from published ARM benchmarks [17].

E. Regulatory and Standards Alignment

TRISecure V2 is designed to align with relevant biometric and e-voting standards. Full conformance certification is de-



(a) Feature comparison heatmap. TRISecure V2 is the only system supporting all six binary capability flags simultaneously.



(b) EER (left) and authentication latency (right) across systems. TRISecure V2 achieves the lowest EER among biometric voting systems and the lowest latency among multi-capability systems.

Fig. 6: Comparative analysis against five reference systems across capability flags and quantitative metrics.

TABLE XVI: TRISecure V2 deployment configurations by station size. Assumes 40 s voter interaction time; 6 h peak window.

Station size	Terminals	Voters/hr	Peak capacity
Small (≤ 500 voters)	1	90	540
Medium ($\leq 2,000$ voters)	3	270	1,620
Large ($\leq 5,000$ voters)	8	720	4,320

ferred pending a live field trial and a Common Criteria certified NFC reader.

ISO/IEC 19795. Biometric performance metrics (EER, AUC, TAR@FAR) are reported following ISO/IEC 19795-1 [53] metric definitions. The simulation methodology follows ISO/IEC 19795-6 [51] operational evaluation guidance, which explicitly permits simulation when field data are unavailable. Full ISO/IEC 19795-1 conformance (Clause 8.5) requires at least 400 subjects per demographic group across gender, age, and ethnicity strata—identified as the primary certification gap.

ISO/IEC 30107-3. PAD metrics (APCER, BPCER, ACER) and attack type taxonomy follow ISO/IEC 30107-3 [24]. Certification evaluation requires standardised corpora: NUAA [28], OULU-NPU [29], and CelebA-Spoof [25]. Current APCER figures (5.0% print, 10.0% replay) are derived from published MiniFASNetV2 benchmark rates and are labelled accordingly.

ISO/IEC 24745 (Template Protection). AES-256-GCM per-database encryption with HKDF-derived keys satisfies the ISO/IEC 24745:2022 [44] confidentiality requirement. The renewability requirement (cancelable transforms) is not yet implemented; cancelable biometric representations [43] are identified as future work (Section VII-D).

NIST SP 800-63B. The pipeline satisfies the structural requirements for Authenticator Assurance Level 3 (AAL3) [7]: multi-factor authentication, physical token (NFC card), verifier-impersonation resistance (on-device processing), and phishing resistance. AAL3 formal certification additionally requires a FIDO2-certified or Common Criteria EAL4+ authenticator device; the PN532 SPI module is not CC-certified.

These alignments indicate TRISecure V2 is *certification-ready* at the design level. The three remaining gaps be-

fore formal conformance can be claimed are: (i) a live voter field trial providing the demographic sample required by ISO/IEC 19795-1; (ii) a certified NFC reader satisfying CC EAL4+ and SP 800-63B AAL3; and (iii) cancelable biometric transforms satisfying ISO/IEC 24745 renewability.

VIII. CONCLUSION

We presented TRISecure V2, a three-factor biometric e-voting system combining formally verified authentication, Bayesian multimodal fusion, ISO/IEC 30107-3 PAD, hybrid post-quantum cryptography, and Merkle-proof voter verifiability—implemented and evaluated on a Raspberry Pi 4 ARM64 platform at under \$100.

The system achieves $EER = 1.55\%$, $AUC = 0.9991$, 110.7 ms end-to-end authentication latency ($p99 = 111.3$ ms), and 541.6 voters/min throughput. Six security properties are formally proved via DFA analysis; three end-to-end verifiability properties are machine-verified. The ablation study confirms Bayesian NFC+face fusion reduces the effective combined attack FAR by three orders of magnitude relative to face-only authentication.

No prior work we are aware of jointly provides formal protocol verification, quantum-resistant cryptography, liveness detection, and edge deployability on sub-\$100 hardware.

Future work includes: live field trial with demographic analysis per ISO/IEC 19795-1 Clause 8; native Kyber-768/Dilithium-3 benchmarks after `liboqs-python` ARM64 packages become available; PAD evaluation on NUAA, OULU-NPU, and CelebA-Spoof; distributed threshold signing; and power profiling for renewable-energy-powered polling scenarios.

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